

2024



Hisense TV System Caters to Global Users with AI-Powered Personalized Recommendation System

Learn how Hisense TV developed an AI-driven intelligent recommendation system on AWS in 1.5 months, improving user experience and driving global expansion.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

1.5 months

to globally launch AI-intelligent recommendation system

Scalability

based on demand

Enhanced Accuracy

Improved TV and movie recommendatic



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Overview



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[Hisense TV System](#), founded in March 2019 by Hisense Group and other investors, focuses on developing smart TV operating systems and content platforms. As the company expanded globally with 20 million worldwide at the end of 2023, addressing the diverse needs of overseas users became crucial.



To enhance the user experience and cater to regional differences, Hisense TV collaborated with AWS to rapidly iterate and launch an intelligent recommendation system tailored for its international users. The system improved recommendation accuracy, optimized user profiles, and supported business expansion in advertising and customer engagement.

Opportunity | Leveraging AI and Cloud Technologies to Address Regional and Cultural Differences in TV Usage

[Hisense TV System](#) focuses primarily on serving overseas TV users by offering a smart TV operating system and content platform. In collaboration with Amazon Web Services (AWS), Hisense TV has expanded its global reach using the AWS Cloud. As the company’s overseas markets and user base grew rapidly, Hisense TV’s product and R&D teams identified significant differences between domestic and international users, including regional cultures, content preferences, and viewing habits.

To better understand these diverse needs and improve user experiences, Hisense TV System’s product and R&D teams began exploring AI and cloud-based innovations. Developing an intelligent recommendation system specifically tailored for international users, and deploying it quickly, became a key priority. The development team, led by Hisense’s R&D Center, was quickly assembled to meet this need. However, given the relatively new application of AI technology, there were few mature references available. Additionally, Hisense’s team, having primarily focused on speech research, had limited AI engineering experience. Addressing this gap in expertise and finding relevant industry references became urgent. Furthermore, the system needed to launch in the European market first to support rapid regional expansion, increasing the urgency.

Solution | Delivering Personalized TV Show Recommendations with an End-to-End Intelligent Recommendation System

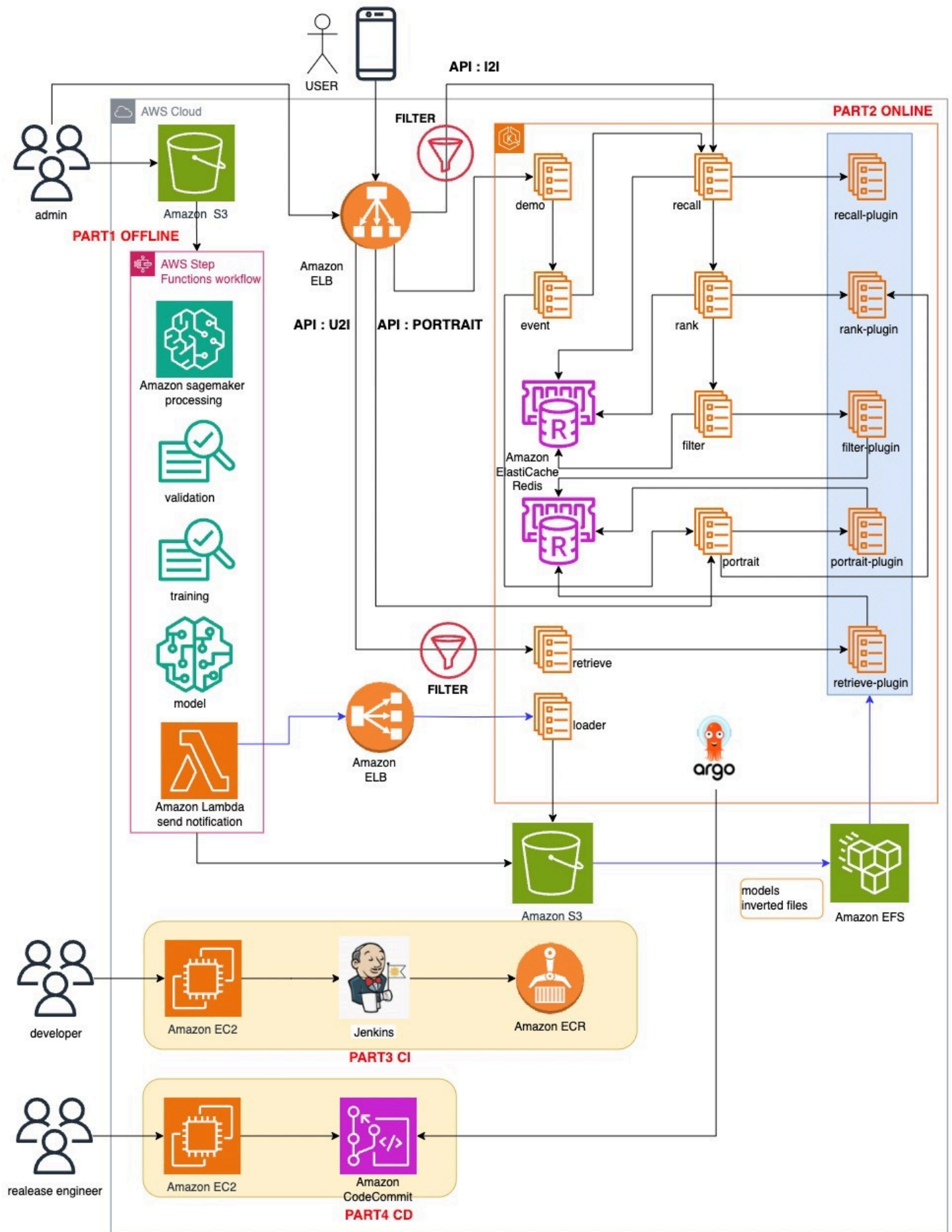
Facing the challenges of a tight timeline and a heavy workload and addressing Hisense TV’s technical and engineering experience, AWS mobilized internal co-creation product experts from its Rapid Product Development team to collaborate closely with Hisense’s technical team. AWS experts were involved throughout the solution design and



system-building process. Together, they built an end-to-end intelligent recommendation system for overseas markets using AWS services such as [Amazon SageMaker](#), [AWS Step Functions](#), [Amazon Elastic File System \(Amazon EFS\)](#), [Amazon Elastic Kubernetes Service](#), [Amazon S3](#), and [Amazon ElastiCache](#).

Amazon SageMaker handled AI model training and deployment, while AWS Step Functions developed data processing workflows. Other services provided storage, compute power, and caching capabilities. The system implemented features like user profiling, content retrieval, and ranking and filtering, providing users with personalized video recommendations. These recommendations were integrated into Hisense TV's video listings, enhancing user experience through small but impactful improvements.

Architecture Diagram



[Click to enlarge](#)

The intelligent recommendation system offers key advantages, including cost-effectiveness, scalability, and fault tolerance. The system builds a user profile library that caters to a wide range of tastes. Powered by robust data processing capabilities, it expands the existing user profile library in a secure and compliant way. In addition to supporting the recommendation system, this library can be efficiently used by other systems or modules to provide personalized services and optimize user experiences.

The system's modular design allows for easy replacement and upgrades. Its microservice-based architecture enables functionality updates, including algorithm changes, with minimal code alterations. This flexibility supports diverse recommendation needs through different API calls. The system also features an elastic architecture, based on container orchestration, which allows it to scale up or down according to demand. This approach optimizes costs while maintaining high availability.

According to Michael Zhu, technical director at Hisense TV Intelligent Research Institute, "The intelligent recommendation system is a key project for 2024. Working with the AWS China team helped us strengthen our AI innovation capabilities. AWS's technical expertise and industry knowledge have been invaluable, and we are confident in the project's success and our ongoing collaboration."

In just 1.5 months, Hisense TV's intelligent recommendation system moved from inception to testing and launch. The system enables automated training and inference of personalized recommendation models, improving the accuracy of content recommendations for TV shows and movies and enhancing the overall user experience. This development is expected to help Hisense achieve its 2024 product goals in the Americas and European markets. The system was officially launched in September as part of the 2024 release of Hisense TV's platform. Now live, the system continues to collect user profile data, which further trains and iterates the models, improving recommendation accuracy over time.

This project also positions Hisense to expand its business in advertising, marketing, and customer service. The system's ability to deliver customized ads, outreach, and personalized customer service based on user preferences is expected to increase user engagement and loyalty.

The successful implementation of this project represents a significant milestone for Hisense in the smart TV operating system and content platform space. By improving the experience of international users, the project supports Hisense's global business expansion while demonstrating the company's ability to rapidly iterate and innovate. This capability will be critical for maintaining a competitive edge in the global market moving forward.

About Hisense TV System

Hisense TV System, founded in March 2019 by Hisense Group and other investors, is an innovative company with internet platform development and operations capabilities.

AWS Services Used

Amazon SageMaker



Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

AWS Step Functions

Step Functions is a visual workflow service that helps developers use AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and machine learning (ML) pipelines.

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Amazon Elastic File System

Amazon Elastic File System (Amazon EFS) automatically grows and shrinks as you add and remove files with no need for management or provisioning.

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Amazon Elastic Kubernetes Service

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

[Learn more »](#)

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2

